Interacting Multipoles of Rank 1 and 2:

$$T_{\alpha\beta\gamma}\mu_\alpha\Theta_{\beta\gamma} = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha \cdot \Theta_{\beta\gamma} \\ +3 \cdot R_z^2 \cdot R_\alpha \cdot \delta(\beta,\gamma) \cdot \mu_\alpha \cdot \Theta_{\beta\gamma} \\ +3 \cdot R_z^2 \cdot R_\beta \cdot \delta(\alpha,\gamma) \cdot \mu_\alpha \cdot \Theta_{\beta\gamma} \\ +3 \cdot R_z^2 \cdot R_\gamma \cdot \delta(\alpha,\beta) \cdot \mu_\alpha \cdot \Theta_{\beta\gamma} \end{bmatrix} \tag{1}$$

Simplify deltas:

$$T_{\alpha\beta\gamma}\mu_\alpha\Theta_{\beta\gamma} = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha \cdot \Theta_{\beta\gamma} \\ +3 \cdot R_z^2 \cdot R_\alpha \cdot \delta(\beta,\beta) \cdot \mu_\alpha \cdot \Theta_{\beta\beta} \\ +3 \cdot R_z^2 \cdot R_\beta \cdot \delta(\alpha,\alpha) \cdot \mu_\alpha \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^2 \cdot R_\gamma \cdot \delta(\alpha,\alpha) \cdot \mu_\alpha \cdot \Theta_{\alpha\gamma} \end{bmatrix} \tag{2a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha \cdot \Theta_{\beta\gamma} \\ +3 \cdot R_z^2 \cdot R_\alpha \cdot \mu_\alpha \cdot \Theta_{\beta\beta} \\ +3 \cdot R_z^2 \cdot R_\beta \cdot \mu_\alpha \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^2 \cdot R_\gamma \cdot \mu_\alpha \cdot \Theta_{\alpha\gamma} \end{bmatrix} \tag{2b}$$

Simplify R:

$$T_{\alpha\beta\gamma}\mu_\alpha\Theta_{\beta\gamma} = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Theta_{zz} \\ +3 \cdot R_z^2 \cdot R_z \cdot \mu_z \cdot \Theta_{\beta\beta} \\ +3 \cdot R_z^2 \cdot R_z \cdot \mu_\alpha \cdot \Theta_{z\alpha} \\ +3 \cdot R_z^2 \cdot R_z \cdot \mu_\alpha \cdot \Theta_{z\alpha} \end{bmatrix} \tag{3a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} +3 \cdot R_z^3 \cdot \mu_\alpha \cdot \Theta_{z\alpha} \\ +3 \cdot R_z^3 \cdot \mu_\alpha \cdot \Theta_{z\alpha} \end{bmatrix} \tag{3b}$$

$$\xRightarrow{\text{added up}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} +6 \cdot R_z^3 \cdot \mu_\alpha \cdot \Theta_{z\alpha} \end{bmatrix} \tag{3c}$$

Simplify Dipoles:

$$T_{\alpha\beta\gamma}\mu_\alpha\Theta_{\beta\gamma} = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} +6 \cdot R_z^3 \cdot \mu_y \cdot \Theta_{yz} \end{bmatrix} \tag{4a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-7} \cdot -\frac{1}{3} \cdot [+0] \tag{4b}$$

Combining everything:

$$T_{\alpha\beta\gamma}\mu_\alpha\Theta_{\beta\gamma} = +0 \tag{5}$$